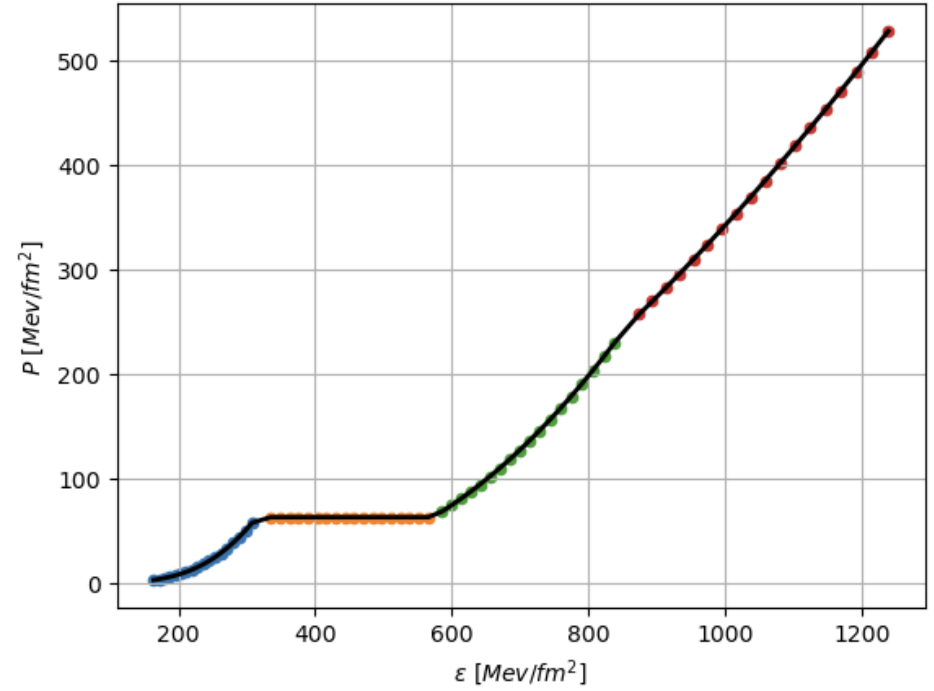
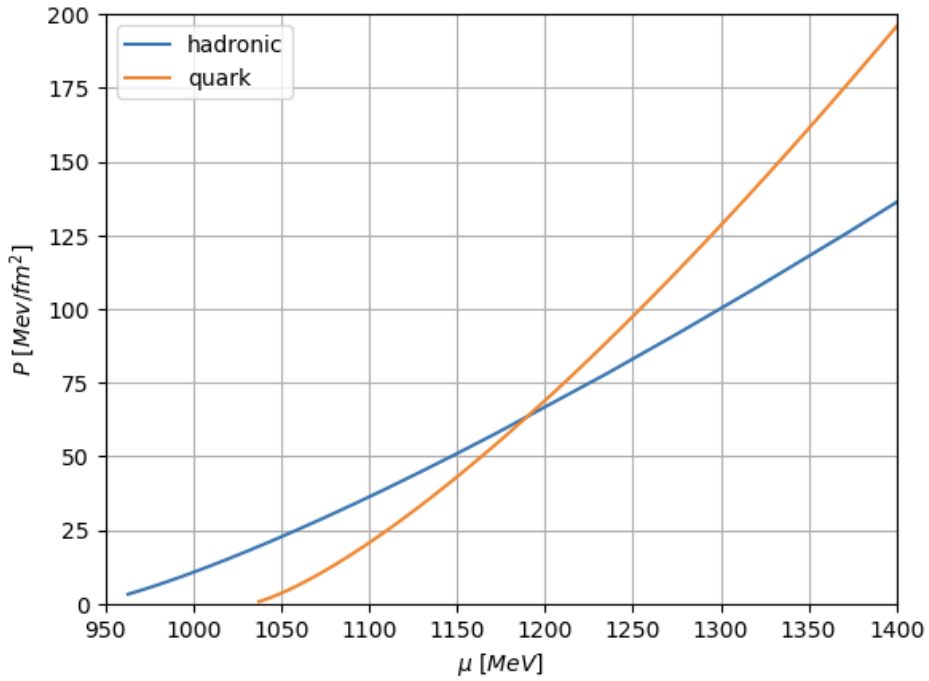
INVESTIGATING MAX PRESSURE JUMP IN MIXED PHASE

Katta Naveen

Supervisor : Dr .Ritam Mallick



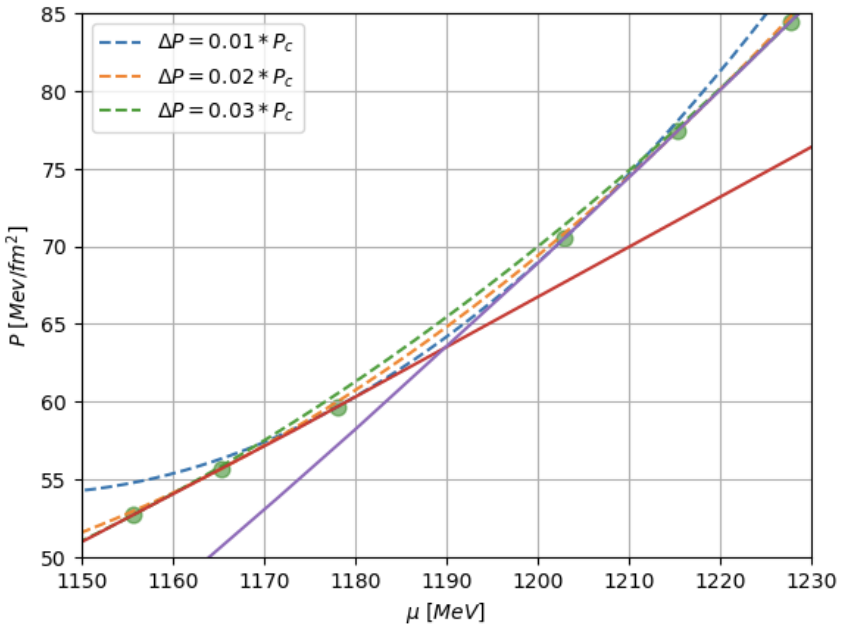


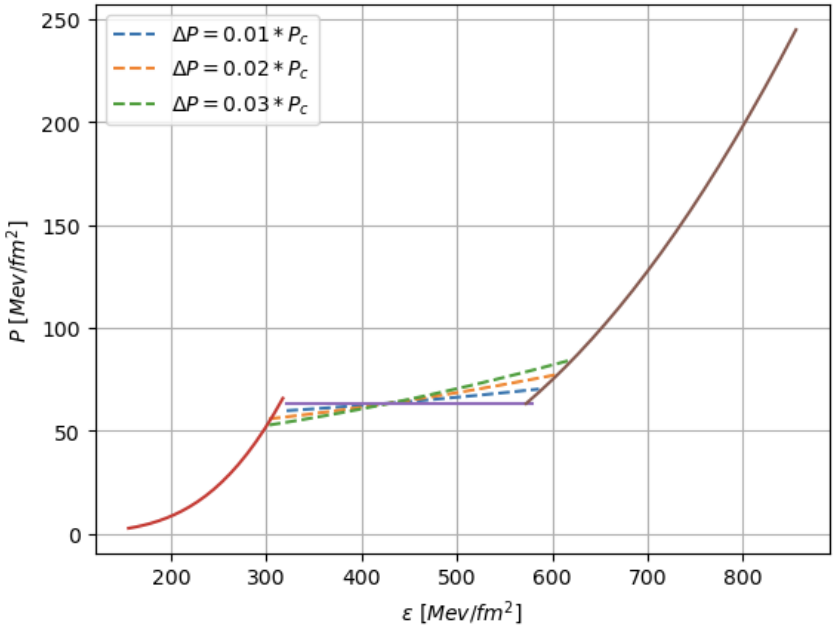


INTRODUCTION

- In neutron stars, extreme high-density and low-pressure conditions—unachievable on Earth—allow us to study phase transitions at those conditions.
- We investigate the phase transition between hadronic to quark matter in the core of neutron stars by using a quadratic interpolation method.
- By implementing a piecewise polytropic equation of state (EOS) for the hadronic and quark phases, we create a smooth transition across the mixed phase.
- We are trying to learn how the smoothness of the transition behaves if we keep on changing the EoS.
- And finally for a any given EoS can we determine the maximum smoothness of the mixed phase.
- We construct EoS using piecewise polytrope.
- The EoS is divided into three polytropes for having hadronic, mixed and quark phase.

QUADRATIC INTERPOLATION

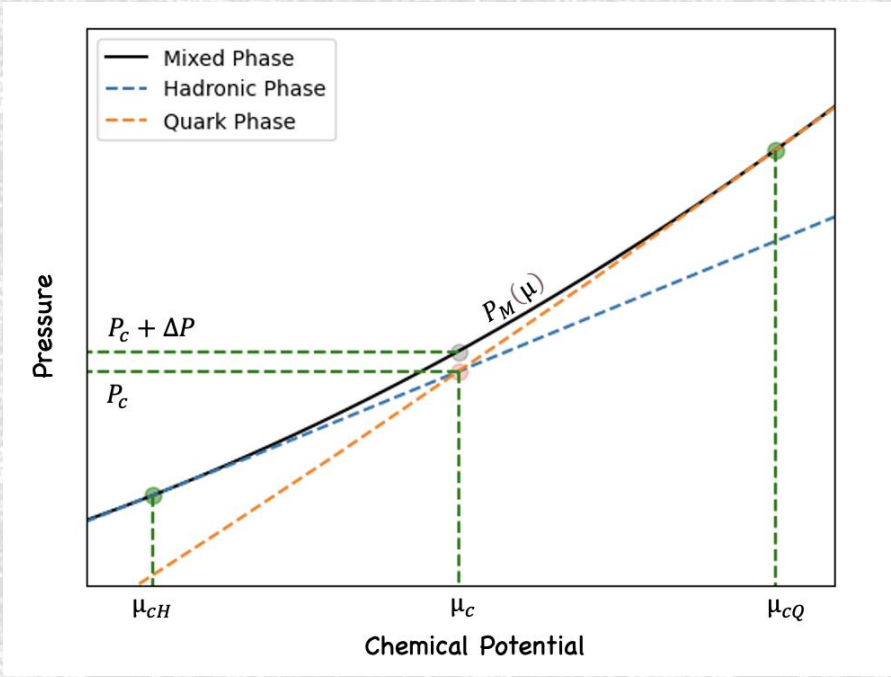


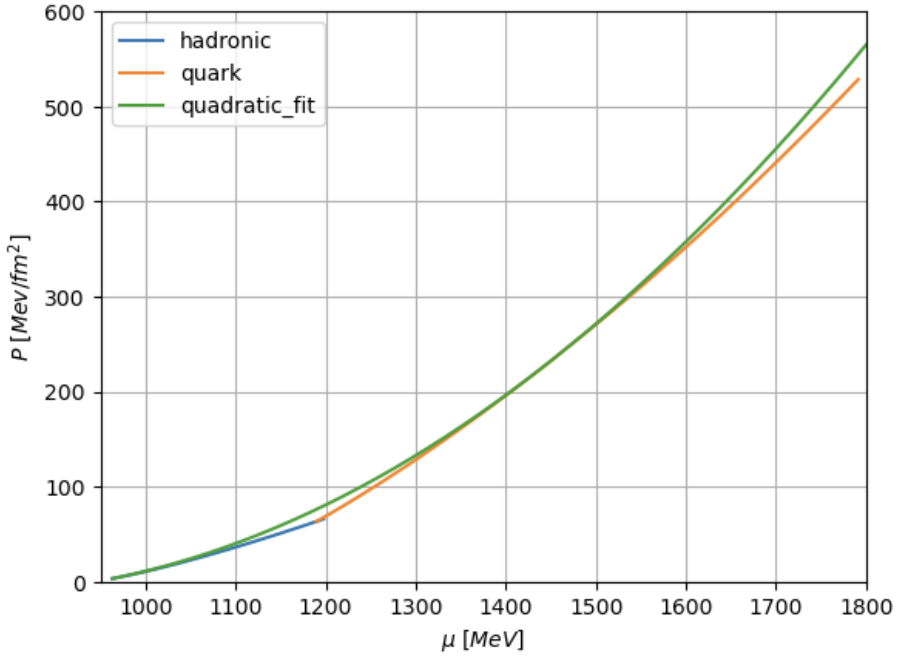


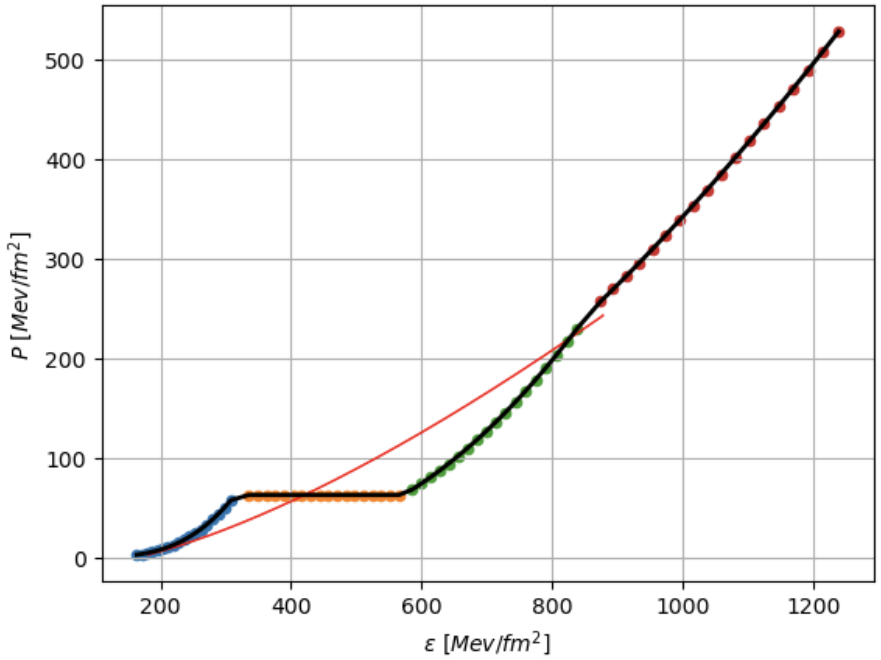
- We consider quadratic form of the mixed phase around a pressure point,
- $P_M(\mu) = \alpha(\mu - \mu_c)^2 + \beta(\mu - \mu_c) + P_c + \Delta P.$
- We have four unknowns : alpha, beta, μ_{cH} and μ_{cQ} .

Boundary Conditions :

- $P_M(\mu_{cH}) = P_H(\mu_{cH}), P_M(\mu_{cQ}) = P_Q(\mu_{cQ}),$
- $n_M(\mu_{cH}) = n_H(\mu_{cH}), n_M(\mu_{cQ}) = n_Q(\mu_{cQ})$

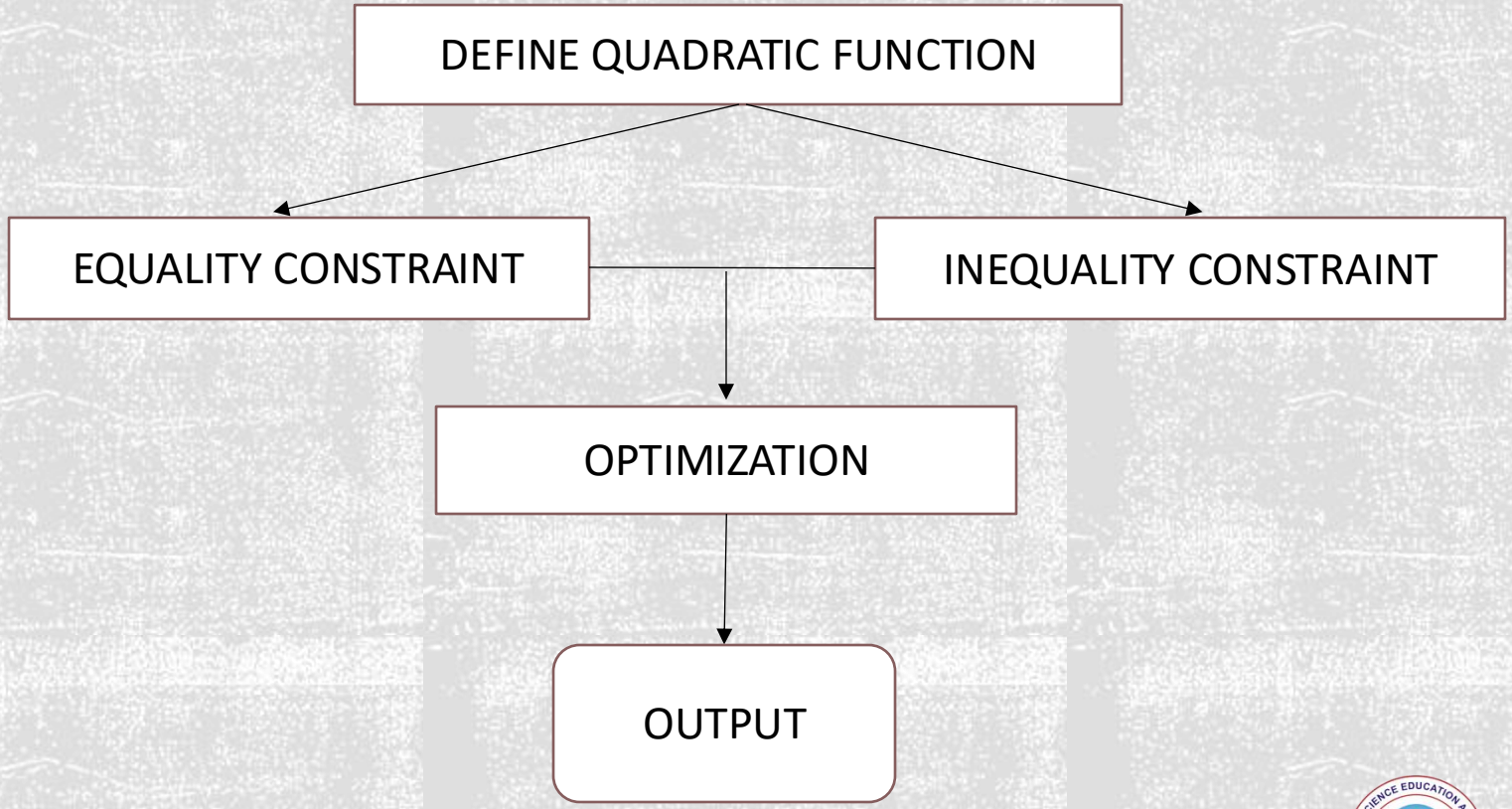


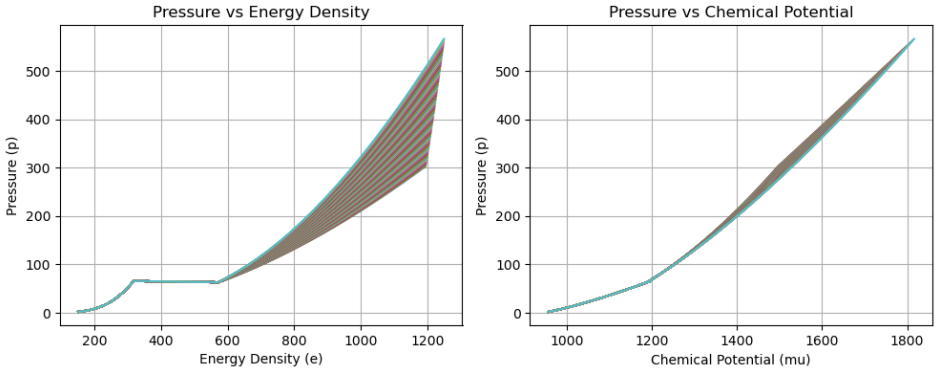


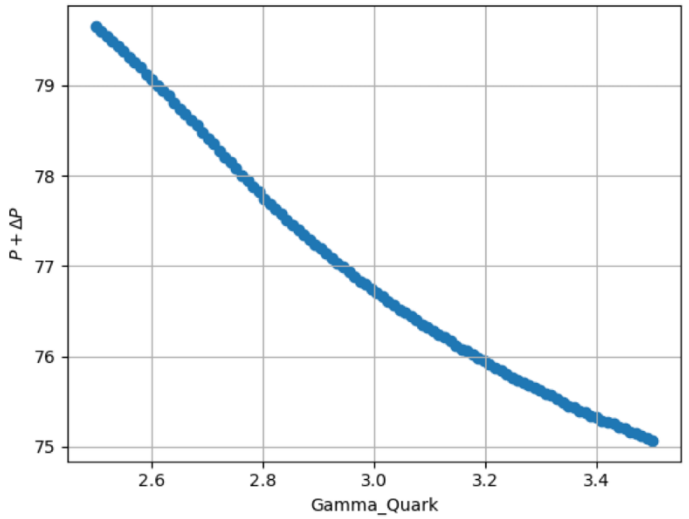


MAXIMUM INTERPOLATION ACHIEVABLE

- Now we will keep the initial transition chemical potential to initial chemical potential of hadronic phase.
- We develop a model such that it finds the optimal values of alpha and beta for such EoS.
- Constraints of the Model : Equality constraint and Inequality constraint.







IMPACT OF DIFFERENT POLYTROPIC INDEX

- If we change the Gamma value of Quark, Polytrope changes so does the EoS, Gamma Quark is ranging from 2.5 to 3.5 .
- Now if we find the maximum interpolation for these different EoS, we get the respective additional pressure terms.
- So in the figure we can see that if we keep on increasing the polytrope index of quark the additional pressure term is keep on decreasing following a trendline.

Future Work

- We want to change the transition point i.e, Pressure and Chemical Potential at critical point, where the transition happens, and observe how the additional pressure changes with respect to these EoS.
- Also match these results with the observational results of Neutron Stars.



THANK YOU